

Lakshya JEE 2.0 (2024)

Mathematics

DPP: 3

Probability

- Q1** The parameters n and p of a B.D are 16 and $\frac{1}{2}$ respectively, then its S.D. is equal to
 (A) 2
 (B) $\sqrt{2}$
 (C) $2\sqrt{2}$
 (D) 4
- Q2** A number is selected at random from the first twenty-five natural numbers. If it is a composite number, then it is divided by 5. But if it is not a composite number, it is divided by 2. The probability that there will be no remainder in the division is
 (A) $\frac{11}{30}$
 (B) 0.4
 (C) 0.2
 (D) none of these
- Q3** A fair coin is tossed a fixed number of times. If the probability of getting seven heads is equal to that of getting nine heads, the probability of getting two heads is
 (A) $\frac{15}{2^8}$
 (B) $\frac{2}{15}$
 (C) $\frac{15}{2^{13}}$
 (D) none of these
- Q4** The mean and variance of a binomial distribution are 6 and 4. The parameter n is
 (A) 18
 (B) 12
 (C) 10
 (D) 9
- Q5** A box contains N coins, of which m are fair and the rest are biased. The probability of getting head when a fair coin is tossed is $\frac{1}{2}$, while $\frac{2}{3}$ when a biased coin is tossed. A coin is drawn from the box at random and is tossed twice. The first time it shows head and the second time it shows tail. The probability that the coin drawn is fair, is
 (A) $\frac{5m}{m+8N}$
 (B) $\frac{3m}{m+8N}$
 (C) $\frac{7m}{m+8N}$
 (D) $\frac{9m}{m+8N}$
- Q6** An experiment has 10 equally likely outcomes. Let A and B be two non empty events of the experiment. If A consists of 4 outcomes, the number of outcomes that B must have so that A and B are independent, is
 (A) 2, 4 or 8
 (B) 3, 6 or 9
 (C) 4 or 8
 (D) 5 or 10
- Q7** Given two events A and B . If odds against A are as 2 : 1 and those in favour of $A \cup B$ are as 3 : 1, then
 (A) $\frac{1}{2} \leq P(B) \leq \frac{3}{4}$
 (B) $\frac{5}{12} \leq P(B) \leq \frac{3}{4}$
 (C) $\frac{1}{4} \leq P(B) \leq \frac{3}{5}$
 (D) None of these
- Q8** From a group of 10 persons consisting of 5 lawyers, 3 doctors and 2 engineers, four persons are selected at random. The probability that the selection contains at least one of each category is



- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$
 (C) $\frac{2}{3}$ (D) None of these

- Q9** The probability of a man hitting a target is $\frac{1}{10}$. The least number of shots required, so that the probability of his hitting the target at least once is greater than $\frac{1}{4}$, is
- Q10** A card is drawn & replaced in an ordinary pack of 52 playing cards. Minimum number of times must a card be drawn so that there is atleast an even chance of drawing a heart, is
 (A) 2 (B) 3
 (C) 4 (D) More than four
- Q11** A bag contains 5 black and 4 white balls. Two balls are drawn. The probability that the balls are of same colour is
 (A) $\frac{4}{9}$
 (B) $\frac{8}{9}$
 (C) $\frac{2}{9}$
 (D) $\frac{1}{9}$
- Q12** It is given that the events A and B are such that $P(A) = \frac{1}{4}$, $P(A|B) = \frac{1}{2}$ and $P(B|A) = \frac{2}{3}$ (Then $P(B)$ is)
 (A) $\frac{1}{6}$
 (B) $\frac{1}{3}$
 (C) $\frac{2}{3}$
 (D) $\frac{1}{2}$



Answer Key

Q1 (A)

Q2 (C)

Q3 (C)

Q4 (A)

Q5 (D)

Q6 (D)

Q7 (B)

Q8 (A)

Q9 (Select & Submit to view solution)

Q10 (B)

Q11 (A)

Q12 (B)



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